

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		hard water contains magnesium ions / calcium ions / Mg^{2+} / Ca^{2+} (1) swap places with two sodium ions (1) accept replace / exchange for swap must be one reference to ions for full credit	2			2		2
		(ii)		either of following - all sodium ions have been used up - no more sodium ions left	1			1		1
	(c)			$0.00135 / 1.35 \times 10^{-3}$ (2) if answer incorrect award (1) for $M_r \text{CaSO}_4 = 136$ award (1) only if answer not given to three significant figures ecf possible from incorrect M_r		2		2	2	
				Question 7 total	5	2	4	11	2	7